

NIELS EVANT ROBERT

SOFTWARE ENGINEER | SECURITY ENGINEER

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Batu Kawa New Township, Jalan Batu Kawa, 93250 Kuching

PROFILE

Security-focused Full-Stack Developer with expertise in designing resilient, AI-integrated applications and enterprise backend systems. Adept at threat modeling, implementing secure API integrations (JWT, OAuth), and mitigating web application vulnerabilities (such as SQL injection) to ensure complete data integrity. Strong background in Agile development, delivering clean, maintainable, and deeply secure code for fast-paced corporate environments.

KEY COMPETENCIES

Full Stack Development	Cyber Security	Android Studio	Teamwork & Collaboration
Web Development	Secure SDLC (Agile/Scrum)	Visual Studio Code	Adaptability & Flexibility
Mobile Development	Data Encryption & Privacy	Azure DevOps	Critical Thinking
System Architecture	CI/CD Pipeline Security	Firebase & Cloud Tech	Problem Solving
API Design & Integration	OAuth 2.0 & JWT Auth	Postman & API Testing	Communication
Data Synchronization	Role-Based Access (RBAC)	Git & Version Control	Continuous Learning
NLP & OCR Integration	Linux Security (SSH/Firewall)	Figma & Canva (UI/UX)	Project Management
Stacks: MERN, LAMP, C#, C++, Java, PHP, SQL, Bootstrap, Laravel, Flutter, Python, Dart, Angular, Vue.js, TypeScript, Bash/Shell, Node.js, TensorFlow		Cybersecurity: API security, SQL Injection & JWT Tampering Prevention, Penetration Testing, Vulnerability Research, OWASP Top 10, CSRF/XSS Prevention	

LANGUAGES

Malay - Advanced	English - Advanced	Mandarin - Beginner	Iban - Beginner
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EDUCATION

UNIVERSITI MALAYSIA SARAWAK	2022 - PRESENT
Bachelor of Software Engineering (Honours)	CGPA : 3.19
UNIVERSITI TEKNOLOGI MARA	2019 - 2022
Diploma in Computer Science	CGPA : 3.26
SMK SOOK	2013 - 2017
Sijil Pelajaran Malaysia (SPM)	Science Stream

WORK EXPERIENCES

INTERNSHIP (SOFTWARE ENGINEER)

Petroliam Nasional Berhad (PETRONAS)

MAR 2025 - SEPT 2025

Engineered a comprehensive management system supporting both Upstream and Downstream operations, significantly enhancing data visibility and cross-departmental workflows. Functioned within an Agile/Scrum environment, bridging the gap between operational needs and technical execution. Leveraged enterprise-grade tools to build scalable solutions, focusing on system reliability, continuous integration, and high-performance front-end interfaces.

- Actively participated in all phases of the Scrum process, utilizing Azure DevOps for sprint planning, backlog grooming, and real-time issue tracking to ensure on-time delivery.
- Assisted in managing CI/CD pipelines within Azure DevOps, streamlining the deployment process and maintaining high code quality across development and staging environments.
- Utilized C# and Angular to develop robust, scalable software features, ensuring seamless integration between complex back-end logic and modern, responsive UI.
- Partnered with senior engineers and team to translate complex operational "User Stories" into technical deliverables, ensuring the system met the specific needs of diverse business units.

IT TECHNICIAN

Imika Empire Sdn Bhd

JULY - OCT 2024

Optimized branch operational efficiency by aligning technical maintenance with commercial objectives. Managed the full lifecycle of hardware and software troubleshooting to ensure zero system downtime during peak sales periods. Actively bridged the gap between technical support and retail performance by providing expert product consultancy and ensuring all technical assets were fully leveraged to drive customer conversion and branch revenue.

- Managed all technical infrastructure and equipment reliability to ensure seamless operations and maximum sales opportunities.
- Sustained branch sales targets by integrating technical troubleshooting with active sales support, ensuring high performance in both technical and commercial scopes.

BUSINESS SYSTEM ANALYST

Flex Sdn Bhd

MAR - SEP 2021

Engineered and optimized Computer-Integrated Manufacturing (CIM) systems to support high-volume production for global industry leaders. Focused on synchronizing complex manufacturing processes with real-time data to drive operational excellence. Ensured the continuous integrity of mission-critical systems and databases within a high-pressure, 24/7 manufacturing environment, minimizing latency and preventing production bottlenecks for multi-national clients.

- Developed and deployed advanced CIM systems tailored for high-profile brands, including Hewlett Packard (HP) and Dyson, ensuring full alignment with global manufacturing standards
- Managed and maintained large-scale databases and system architectures to support non-stop, 24-hour production cycles, guaranteeing 99.9% system uptime.

INTERNSHIP (FULL STACK DEVELOPER)

Sarawak Information System Sdn Bhd (SAINS)

NOV 2020 - FEB 2021

Contributed to the digital transformation of internal organizational workflows by developing robust web applications tailored for management use. Engaged in the full project lifecycle—from initial requirements gathering and UI/UX design to backend implementation and final deployment. Successfully bridged technical requirements with user needs to deliver scalable, high-performance web solutions that improved administrative efficiency and data visibility.

- Engineered a comprehensive management-focused website, prioritizing clean architecture and a user-centric interface to streamline internal reporting and operations.
- Managed the development of assigned software projects from the conceptual phase through to a successful production release, ensuring all technical milestones and quality standards were met.

FEATURED PROJECT

DEGREE FINAL YEAR PROJECT

OCT 2025 - JULY 2026

MedSense - Intelligent Pill Recognition and Prescription Verification System

An AI-powered mobile application designed to mitigate medication errors by integrating computer vision and natural language processing (NLP). The system allows users to identify pills via camera analysis and cross-references them with stored prescription data to provide real-time safety alerts and drug interaction warnings.

- Frontend: Flutter (Dart)
- Backend: Python (FastAPI)
- AI & Machine Learning: TensorFlow Lite (Image Classification), OCR (Optical Character Recognition), and Natural Language Processing (NLP)
- Databases: PostgreSQL (Cloud storage) and SQLite (Local offline storage).
- Tools & Technologies: Firebase (Cloud Messaging/Authentication), Postman, Git, Figma (UI/UX Design), and Android Studio
- Development Methodology: Rapid Application Development (RAD)

ROADSENSE

DEC 2024 - MAR 2025

Roadsense – Crowdsourced Civic-Tech Infrastructure Monitoring System

A civic-technology mobile application designed to autonomously map infrastructure degradation. The system aggregates road health telemetry by processing high-frequency sensor data and real-time camera feeds to classify pothole severity and optimize routes via geospatial heatmaps.

- Frontend: Flutter (Dart), Google Maps API
- Backend: Cloud-based Data Ingestion Pipeline (Firebase/Node.js), RESTful APIs
- Tools used: Native Device Hardware APIs (Accelerometer/Gyroscope), TensorFlow Lite (Computer Vision), Git, Postman

BACASAMA

OCT 2024 - JULY 2025

BacaSama – Phonics-Based Literacy Learning Platform

Engineered a specialized digital literacy platform designed to improve Malay reading and writing skills for non-technical adult demographics in rural Sarawak. The application features an offline-first architecture with phonics-based lessons, speech recognition, and OCR to ensure continuous accessibility in environments with intermittent connectivity.

- Frontend: Flutter (Dart)
- Backend: Firebase (Authentication, Firestore Database), RESTful APIs
- Tools used: Google ML Kit (OCR & Speech Recognition), RiverPod (State Management), Android Studio, Git

ATLAS BY ACCURATE

JAN 2024 - DEC 2024

Atlas – Digital Land Management & Field Operation System

Architected a digital management system for the Sarawak Land and Survey Department to transition paper-based workflows into an automated environment. Engineered a cross-platform mobile app for real-time field data collection and integrated secure RESTful APIs to ensure the integrity of sensitive land surveying data via role-based access controls (RBAC).

- Frontend: Flutter (Dart), HTML, CSS, Bootstrap 4.5, Ajax, JavaScript
- Backend: RESTful API Design, PHP, MySQL / PostgreSQL
- Tools used: Android Studio, Git, Postman, Figma

ACHIEVEMENTS

UNIVERSITI MALAYSIA SARAWAK

FINALIST - MOBILE APP COMPETITION 8.0

2026

FCSIT UNIMAS

- Engaged in a mobile app competition targeting smart city solutions, aimed at improving urban living through technological innovation.
- Demonstrated expertise in user experience design using figma to propose a viable solution aligned with smart city initiatives.

SPEAKER | EMPOWERING CLASSROOM TEACHING WITH GEMINI

2025

SERVICE LEARNING - FCSIT UNIMAS

- Facilitated live demonstrations of Gemini's multi-modal capabilities, including prompt engineering and AI-driven content generation for educational workflows.
- Engaged with a diverse audience to address the challenges and opportunities of AI in the Malaysian education sector.

PARTICIPANT - NASA INTERNATIONAL SPACE APPS CHALLENGE 2025**2025**

NASA'S EARTH SCIENCE DIVISION

- Actively participated in the renowned global hackathon hosted by NASA, contributing to solving real-world challenges in space exploration and technology.
- Collaborated within a team environment to address specific challenges set forth by NASA, demonstrating problem-solving skills and technical expertise.

FINALIST - OPEN CATEGORY CODENECTION 2023**2024**

MULTIMEDIA UNIVERSITY (MMU), CYBERJAYA

- Engaged in solving challenging coding problems within strict time constraints.
- Advanced to the final round held at RekaScape Cyberjaya, participating in an on-site event.
- Demonstrated abilities in coding, problem-solving, and teamwork.

TOP 7 - IEEEEXTREME 24-HOUR PROGRAMMING COMPETITION**2023**

Institute of Electrical and Electronic Engineers (IEEE)

- Demonstrated the ability to maintain precision and logic throughout a 24-hour continuous development cycle.
- Effectively coordinated with team members to analyze problems, divide tasks, and peer-review code under intense time pressure.

TOP 10 - PROGRAMMING LEAGUE NATIONAL 2023**2023**

UNIVERSITI MALAYA (UM)

- Maintained high accuracy and speed during a multi-hour live coding contest against elite teams from universities across Malaysia.
- Focused on reducing time and space complexity to ensure solutions passed all hidden test cases within the required limits.

RUNNER-UP - BORNEO INTERNATIONAL DRONE AND ROBOTICS EXPLORE (DROBOTEX)**2019**

UNIVERSITI MALAYSIA SARAWAK (UNIMAS)

- Implemented maze-solving algorithms (such as Wall Follower or Flood Fill) to ensure the shortest path calculation and obstacle avoidance.
- Iterated on hardware design and code logic under tight competition timelines to improve speed and reliability.

REFERENCES**TS. NURFAUZA BT JALI**

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